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# Prelab 6.1 Database & Visualization 1 - SQL Database

## Learning Goals

Students will be able to:

1. Execute basic SQL statements and clauses to perform actions on a database
2. Write a Python script to execute SQL statements that modify a SQL database

## 1.1 Introduction

In this lab, we will use a database to store data that is collected by IoT sensors and devices. In Lab 4 and 5, we learned MTConnect as a middleware to integrate data flow. Because data size in the MTConnect agent is limited to the memory size of the computer, introducing database is needed to store data for long-term use.

There are many different types of databases. Normally the types of databases are divided into SQL (Structured Query Language) and NoSQL (non SQL or not only SQL). The selection of a proper database is important when you start a project. Of course, NoSQL can be better in many applications. However, of the various databases, we will use SQL (Structured Query Language) database, specifically MySQL. SQL is a standardized programing "language" that is used to manage relational databases and perform various operations on the data in them. Because SQL database is elementary of database, we will use SQL for the lab activities. SQL allow us to store, update, and retrieve data in databases. MySQL is an open-source (royalty free) relational database management system (RDBMS). It is developed by Oracle Co. MySQL is a representative RDBMS and it is used by many popular websites, including Facebook, Flickr, YouTube and so on.

The purpose of Lab 7 is 1) to understand the basic syntax of SQL and 2) to manipulate (store and retrieve) databases. As a hands-on activity, we will store sensor data into MySQL database using SQL and retrieve data from it.

## 1.2 MySQL Python Package Installation on Raspberry Pi

In Lab 6, we will program a Python script to store sensor data into MySQL database. PyMySQL (https://pypi.org/project/PyMySQL/, available on Feb. 28, 2022) will be used. Please follow the steps to install PyMySQL Python package below.

1. To make your Raspberry Pi up to date and upgrade

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**Raspberry Pi - Terminal**

`sudo apt update`

`sudo apt upgrade`

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2. Install PyMySQL Python package (latest version is 1.0.2)

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**Raspberry Pi - Terminal**

`sudo pip3 install PyMySQL`

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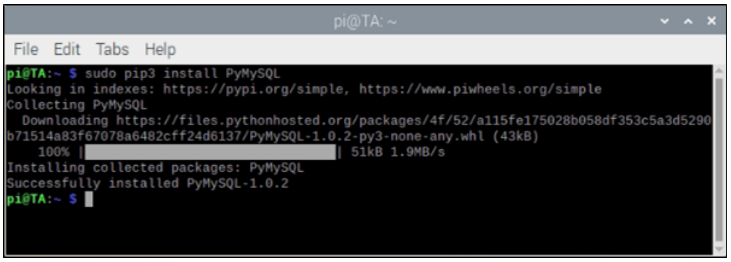


*Figure 1 PyMySQL installation on Raspberry Pi*

3. Check if PyMySQL is installed as below

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**Raspberry Pi - Terminal**

`python3`

**Python - Python3 interpreter in 'Terminal'**

`import pymysql`

`print("PyMySQL version:", pymysql.__version__)`

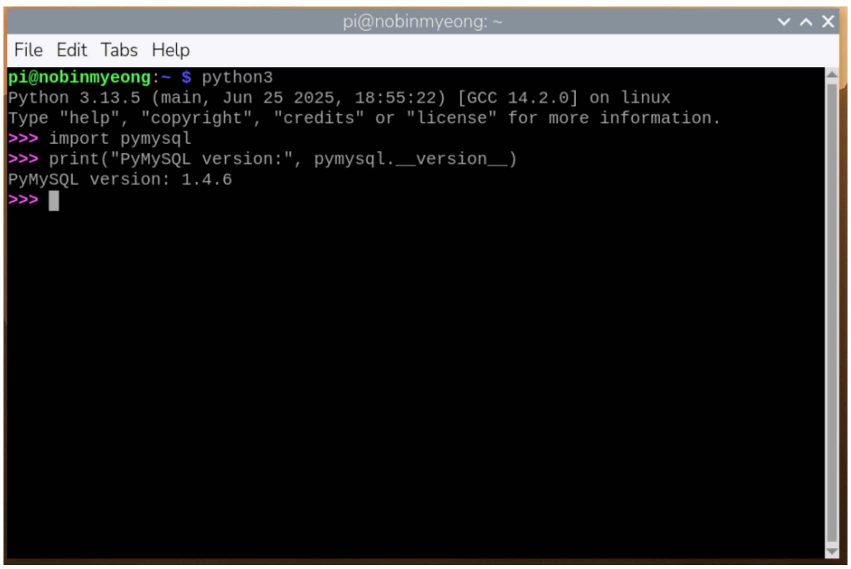


*Figure 2 PyMySQL import and check the version*

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## Task 1.1

Install PyMySQL package on Raspberry Pi. Check the version of PyMySQL and then attach the result of the terminal window as Figure 2 to the report.

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- To exit the Python interpreter, the command is ‘exit()’.

## 1.3 MySQL Workbench Installation on Windows

We can manage MySQL database through command line interface. However, it requires fluent skills on both MySQL client and CLI. We will use MySQL Workbench. MySQL Workbench is a unified visual tool for database architects, developers. It provides data modeling, SQL development, and comprehensive administration tools. We will use MySQL Workbench to practice basic syntax of SQL. To install MySQL Workbench, please follow the steps below.

1. Visit MySQL Community Downloads
2. Download and install MySQL Workbench Community version 8.0.X

a. You can install other versions but please make sure to install the version higher than 8.0.

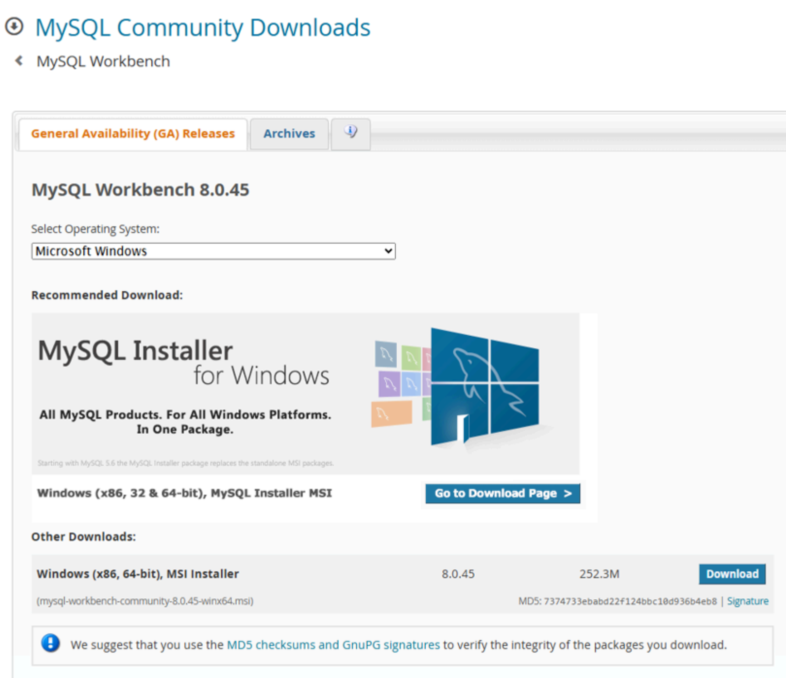
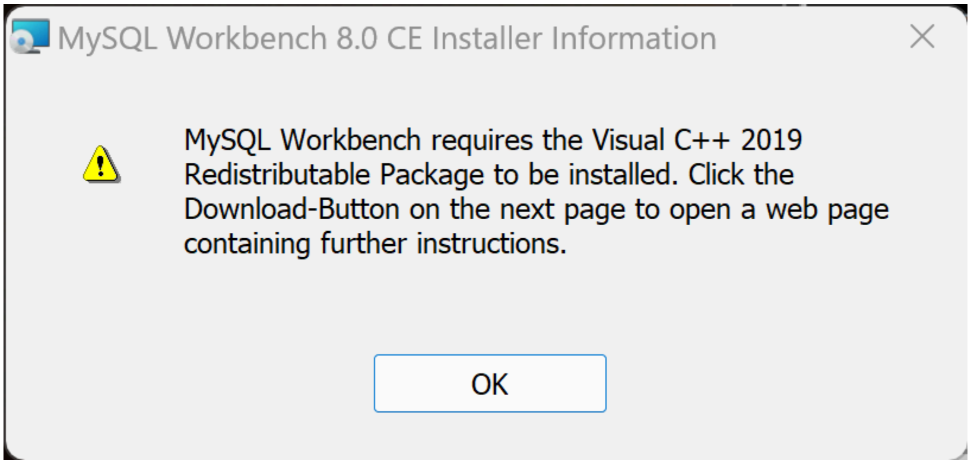


*Figure 3 MySQL Workbench window*

## Troubleshooting

If you encounter an error message while downloading MySQL, it means that the compatible 'Visual C++' software is not installed. Please follow the guidelines and retry the download process.




First, enter the following Visual C++ download link.

https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170

Click the following link, as shown in the red box.


Select a C++ download link that matches your OS version and is compatible.

(x86: Window 32-bit, x64: Window 64-bit, ARM64: Linux 64-bit)

# Latest Microsoft Visual C++ Redistributable Version

The latest version is `14.42.34433.0`

Use the following links to download this version for each supported architecture:

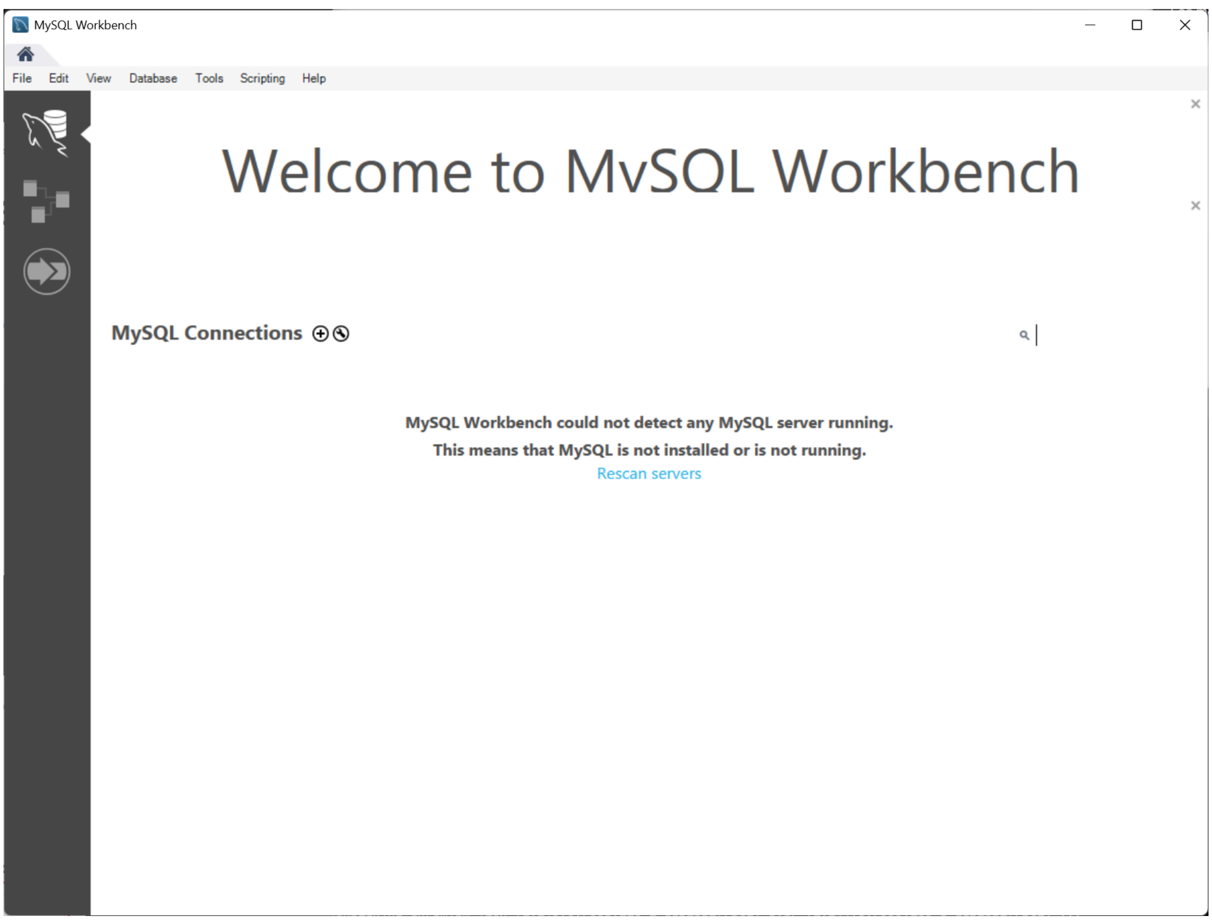
Expand table

| Architecture | Link | Notes |
|---|---|---|
| ARM64 | https://aka.ms/vs/17/release/vc_redist.arm64.exe | Permalink for latest supported ARM64 version |
| X86 | https://aka.ms/vs/17/release/vc_redist.x86.exe | Permalink for latest supported x86 version |
| X64 | https://aka.ms/vs/17/release/vc_redist.x64.exe | Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when the X64 Redistributable is installed on an ARM64 device. |

Download other versions, including long term servicing release channel (LTSC) versions, from my.visualstudio.com .

## Task 1.2

Install MySQL Workbench on Windows and run it. Capture the window as Figure 3 and then attach it to the report.



## Task 1.3

Complete the SQL tutorial in the following URL from the section 'SQL Home' to the section 'SQL Like'.

- (Tutorial) W3Schools: https://www.w3schools.com/sql/default.asp

## Task 1.4 Answer the following questions

### 1. Describe the structure of a table and the components that make up a SQL table.

A table is defined by a name and contains records or rows of data. SQL tables are formatted similarly to excel or .csv tables in that they are tabular in the format of columns and rows. SQL tables differ in that tables can be relational to one another. Several tables can all relate to one another even when they are different sizes and contain different data.

### 2. Explain the function of each of the SQL statemnts or clauses below.

T4_SELECT: SELECT is the statement in which you choose which column to pull data from. T4_INSERT_INTO: INSERT_INTO is the statement used to add more data to a table in a SQL database. T4_WHERE: WHERE is a clause used to add conditionals to filter out certain parts of the data retrieved from the database. T4_ORDER_BY: ORDER_BY is a clause used to sort the data by one of the columns in a database in ascending or descending order. T4_FROM: FROM is the statement in which you declare the table to pull the selected columns from.

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